Perturbation Equivalence in Closed-Loop Agents: When Is a Compressed Agent Actually Worse?

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Abstract

Compressed LLM agents are usually judged against a single run of the full-precision model. We argue that in closed-loop evaluation under greedy decoding this comparison is ill-posed. Composing a published per-step margin model of greedy token flips over an episode, we predict, and observe on Qwen3-4B (consistently on Qwen3-8B), that every perturbation whose logit shifts reach the bf16 grid step (0.125 nats) meets a per-decision divergence floor set by near-ties in free-text turns: a prompt nonce, a compiler kernel choice and 8-bit rounding all re-roll the trajectory. We therefore propose *perturbation equivalence*: a compressed agent is judged by task-level exchangeability with outcomes from the reference model's own perturbation null. On τ^2-bench retail and airline (85 tasks), serving numerics alone move the canonical Qwen3 score by 6–8 points and flip about a quarter of task outcomes, and a single-draw 12-point calibration gain reverses sign on replication. On Qwen3-4B, success is not distinguishable from the null up to the RTN-W4 error scale and drops only near 3 bits, a tentative later damage threshold. [PENDING: five-model summary sentence — cells `a6k2_{q4,q8,q14,q30a3,q32}_*`, scripts `analysis/insights/iter_1/verify/v01_legacy_recompute.py` a6k2, `v03_outcomes_pooled.py` a6k2, `v04_pooled_draws.py`] We close with an evaluator checklist.

1 Introduction

Agent benchmarks such as τ^2-bench (Barres et al., 2025) score a model by letting it interact with a simulated user and a tool environment until the episode ends, and then checking the final database state. When a compressed model is evaluated this way, the usual report is a pair of numbers: the score of the BF16 reference and the score of the compressed model, each from one run. We ask what such a pair can tell us, a question about evaluation methods for agentic systems rather than about any one compression method.

Two observations from our own project are warnings. First, in an early study of calibration data for 4-bit GPTQ on Qwen3-4B, a trajectory-based calibration arm (FP-Traj) beat a static arm by $+0.121$ in database match on 58 retail tasks (8 tasks won by FP-Traj alone, 1 by Static alone; exact one-sided McNemar $p = 0.0195$, two-sided $p = 0.039$); the replication with three independent calibration draws gave -0.035 pooled (-0.045 on long tasks), with the sign reversed and both intervals spanning zero. Second, and sharper because the model did not change at all: re-serving the same BF16 weights with the same prompt under a different floating-point configuration of the serving stack (for agent and simulator) moved the canonical score of Qwen3-4B from 0.412 to 0.329 and that of Qwen3-8B from 0.316 to 0.380, changing 21 of 85 and 21 of 79 task outcomes. A compressed model compared against either reference run would receive a different verdict.

We explain both observations with one mechanism. Greedy decoding picks the top token at every position. When the top two candidates are nearly tied, any perturbation of the logits that exceeds the gap can flip the choice. This per-step account is established (Du et al., 2026; Chen, 2026; Chu et al., 2026). In a closed-loop agent, a flipped token changes the context of every later turn, including the simulated user's replies, so a single flip *re-rolls* the rest of the trajectory. Free-text turns contain many near-ties, so a perturbation of almost any size re-rolls almost every episode. The outcome of each task is then better described as one Bernoulli draw than as a fixed property of the model.

The question "is the compressed agent worse?" must then be answered against the outcomes the reference model itself produces under perturbations that carry no capability change: prompt nonces, tiny weight noise, and serving-numerics changes. We make three contributions.

1. **A criterion.** We define *perturbation equivalence*: a compressed agent is equivalent to its reference if its task-level outcomes are exchangeable with those of a null family built from the reference model's own sub-threshold perturbations. The canonical reference score is one draw from this family (serving numerics alone flip about a quarter of task outcomes; null-draw SDs are consistent with a Bernoulli prediction), so single-draw comparisons can produce false positives such as the one above.

2. **A closed-loop composition of the per-step margin model, with two measured regimes.** The composition predicts (P1) a per-decision divergence *floor* reached by every perturbation at or above the bf16 grid step, then a regime where the hazard ranks perturbation size; (P2) a *damage* threshold later than the divergence threshold, leaving a zone of full re-rolls with success within the null; and (P3) divergence predictable from the reference margins. P1 holds on Qwen3-4B on clean data and is consistent on Qwen3-8B; P2 is a tentative finding on Qwen3-4B; P3 is partly supported and partly at risk.

3. **A serving-numerics finding and a deterministic serving recipe.** A compile-time autotuning choice in the fused residual + RMSNorm kernel changed the summation order across nodes, shifted all 113 identical-prompt decisions of a determinism pair (non-zero shifts on 59% of tokens), and re-rolled about 95% of its trajectories (Kaplan–Meier estimate over 20 tasks). The fix (deterministic Inductor configuration plus a numerics canary) restores cross-node byte-identity (257/257 and 303/303 decisions).

An earlier 4B study (Appendix E; two RTX 4080 GPUs, 8k harness, the 4B model as its own simulator, numbers recomputed from analysis outputs) adds that 18 GPTQ checkpoints from six calibration strategies are statistically indistinguishable in closed loop (arm-label permutation $p = 0.683$), and that teacher-forced NLL does not track closed-loop success.

2 Related Work

Compressed LLMs as agents. ACBench (Dong et al., 2025) evaluates 15 compressed models at temperature 0 (tool use drops 1–3% at 4 bits, real-world applications 10–15%); our notes record one run per configuration and no reported variance. The closest work is *Flat Score, Amplified Failures* (Jang et al., 2026): BF16, FP8 and AWQ-INT4 agents on τ^2-bench retail and telecom with four trials per task. It finds no score change surviving multiple-comparison correction, reports TOST equivalence at ± 7.5 points with a noise floor (about 1.5 points) from one repeat run, and shows failure modes rising up to $2.5\times$ while the aggregate stays flat; a two-error budget re-exposes a 17-point gap. Its margin account of flipped decisions is verbal (token log-probabilities are not in its logs), and it measures neither divergence nor a perturbation null; we add both, plus measured margins (Table 4, Appendix C). Other work reports token inflation under low-bit quantization (Lian et al., 2026), task-dependent precision needs of quantized agents (Lu et al., 2026; Zhang et al., 2026), function calling robust to aggressive quantization (Taheri et al., 2026), and closed-loop evaluation of quantized vision-language-action policies (Xu et al., 2026).

Evaluation noise and reference nulls. See Heineman et al. (2025); Kapoor et al. (2024); Yao et al. (2024) for benchmark separability, agent benchmarking practice and pass$\hat{}k$ reliability. On τ^2-bench, an analysis of released trajectories estimates that single runs must differ by 7–8 (retail) or 10–11 (airline) points to exceed noise under sampling (Universeyi, 2026); single-run pass@1 of SWE-bench Verified agents varies by 2.2–6.0 points, with SDs above 1.5 points even at temperature 0 (Bjarnason et al., 2026). On single-turn BFCL, prompt perturbations produce paired SDs 11–58$\times$ those of reruns, with multi-turn agents out of scope (Chen et al., 2026). Singh (2026) finds per-item churn about $5\times$ the net delta for compressed models on static benchmarks and proposes paired equivalence testing at a *declared* margin (± 2 points in its example). The Replay Gap (Gonuguntla, 2026) uses same-model control forks as a null for model switching in SWE agents; FP8-served forks diverge in over 90% of cases while AWQ-served ones stay near-identical, so rerun-null width depends on the serving format. Statistical precedents include composite perturbation nulls (Acharyya et al., 2025), data-dependent equivalence margins (Koobs and Koning, 2026) and counterfactual-prompting baselines (Yang et al., 2026). Our null comes from the reference model's *own* perturbations, task by task, in a

closed loop, with its width explained by a margin model.

The per-step margin model (inherited). Du et al. (2026) show that BF16 and FP16 greedy decoding diverge on 49–100% of prompts and that a token flips when the perturbation of the logit gap exceeds the top-two margin, with the heuristic $P(\text{flip at } t) \lesssim P(\Delta_t < \sigma_z)$ (their Eq. 14). The same relation holds for INT8 kernel swaps (margin-to-flip ROC-AUC 0.942; Chen, 2026), and low margins can be exploited to restore determinism under batching (Chu et al., 2026); numerical nondeterminism in serving is documented by Yuan et al. (2025) and He and Thinking Machines Lab (2025). We take the per-step model as given; our contribution is its composition over a closed loop, the resulting floor and two-regime structure along one perturbation axis, the later damage threshold, and its use to define and interpret a null.

Divergence in multi-turn settings; open vs. closed loop. Prefix caching changes 36.2% of multi-turn tool episodes at 16 bits and 75.0% at 4 bits, with saturation at the coarsest settings and a 3.5% single-turn-bridge rate explained verbally by compounding (Patodiya, 2026). Meaning-bearing rewrites destabilize agents more than presentation changes (Zhang and Guo, 2026) (our nonce is a presentation change); multi-turn loss is mostly unreliability rather than aptitude (Laban et al., 2025); long-horizon execution compounds errors (Sinha et al., 2026); and classical work on compounding error in sequential prediction (Ross et al., 2011; Bengio et al., 2015) motivates the closed-loop view. Equal accuracy can hide changed predictions (Dutta et al., 2024; Deiseroth et al., 2023), distribution-divergence metrics complement accuracy (Qamar et al., 2026), and in driving, open-loop metrics correlate with closed-loop scores only non-monotonically (Wang et al., 2026). We add a direct open- versus closed-loop measurement for quantized tool agents (Appendix E).

3 Framework

3.1 Setting and notation

An agent M interacts with a user simulator and a tool environment on task t. At decision s it emits a turn (free text or a tool call) token by token under greedy decoding. The task outcome $Y_t(M) \in \{0, 1\}$ is the database end-state match; premature termination is scored as failure.

For each generated token j under the reference, the *margin* m_j is the gap between the top-1 and top-2 log-probabilities. A perturbation π (a prompt nonce, a weight change, or a change of serving numerics) shifts the reference log-probability gap at token j by d_j; D_π is the distribution of these shifts and δ its scale. The Gaussian noise level k (§4) is the *dose* we apply; the median of $|d|$ is the *shift* we measure, either *closed-loop* (unflipped prefix tokens of identical-context decisions, on the bf16 log-probability grid; Table 2) or *open-loop* (a replay over all tokens of the reference contexts; Appendix D). The two are different quantities that we never compare numerically.

3.2 Per-step model (inherited) and closed-loop composition

Following Du et al. (2026), a token flips when the shift overcomes the margin, $P(\text{flip}_j) \approx P(d_j < -m_j)$. A decision s diverges if any of its tokens flips on an identical context,

$$p_s = 1 - \textstyle\prod_{j \in s} P(d_j \geq -m_j). \qquad (1)$$

In a closed loop, a divergence at decision s changes the context of all later decisions, including the simulator's replies, so

$$P(\text{diverged by } t) = 1 - \textstyle\prod_{s \leq t}(1 - p_s), \qquad (2)$$

and p_s is the per-decision divergence hazard. The composition is elementary; what matters is what it implies when margins are structured.

3.3 Predictions

Text turns contain several near-tied tokens per decision, while tool calls are usually decided by margins of several nats (§5.1). This two-tier structure yields three predictions.

P1 (floor, then ranking). Any perturbation whose shifts reach the bf16 log-probability grid step (0.125 nats in our closed-loop measurements) flips text-turn near-ties at a rate set by the density of ties, not by δ, so the per-decision hazard has a floor that every such perturbation reaches and episode-level divergence saturates. Once tool-turn margins start to be crossed, the hazard grows with δ and ranks perturbations. With $D \equiv 0$ there is no divergence.

P2 (damage later than divergence). A re-rolled trajectory is a new sample from roughly the same policy, so expected success should stay within the null until δ reaches the margins of state-changing actions and those flips are not recovered; damage thus starts at a larger δ than divergence, leaving a *free zone*.

P3 (predictability). The perturbation enters the per-step model only through D, so a model's divergence curve should be predictable from its reference margins and D, and perturbations with matched D should have matched closed-loop outcomes. This is a hypothesis; §5.3 reports evidence that may falsify it for RTN.

3.4 Perturbation equivalence

Let $\mathcal{P}_M = \{\pi_1, \ldots, \pi_R\}$ be a null family of perturbations of the reference M that are not intended to change capability; each member yields an outcome vector $Y^{(r)} = (Y_t^{(r)})_{t \in T}$. In this paper the family draws on prompt nonces (a presentation-only session-identifier line in the system prompt), a serving-numerics variant (eager instead of compiled execution), and weight-only Gaussian noise at $k = 0.03$. The pooled null used for testing (§5.2) contains the canonical, nonce and eager runs; the $k = 0.03$ draws are analysed as a separate family, because whether they belong in the null is itself open (§5.2, §5.4).

Definition. A compressed model C is *perturbation-equivalent* to M on task set T if the columns of the task $\times$ run matrix $[Y^{(1)}, \ldots, Y^{(R)}, Y(C)]$ are exchangeable.

Under P1, each task outcome in the null is approximately a Bernoulli draw with task-specific rate p_t, so the run-level standard deviation of the null is

$$\mathrm{SD}_{\mathrm{run}} \approx \sqrt{\tfrac{1}{n}\,\overline{p_t(1-p_t)}}, \qquad (3)$$

where n is the number of tasks and the bar denotes the mean over tasks. We operationalize the criterion with three statistics, in decreasing order of maturity: (i) $\Delta = \frac{1}{n}\sum_t (Y_t(C) - \bar{Y}_t^{\mathrm{null}})$, pooled over domains, with a task-cluster bootstrap CI (pre-registered MDE of the pooled five-model free-zone test: about 3.4 pp); (ii) the *cell-only win rate* (tasks the cell passes that the null majority fails) against the null's own band, which detects nested pass sets (damage) rather than re-rolls; (iii) a permutation test on mean pairwise ϕ between columns (null–null vs. null–cell). [PENDING: ϕ permutation test (P4b) — task × run matrices of a6k2 q4 and q8, analysis E4-f in `analysis/insights/iter_1/verify/v04_pooled_draws.py`]

Work on static benchmarks shows that scores change under paraphrase and across prompt templates, with paraphrased questions often scoring below the original wording (Lunardi et al., 2025; Mizrahi et al., 2024), so a nonce null might sit systematically below the canonical run. Canonical-prompt members (weight-only and numerics) are therefore the preferred null, with nonces reported alongside; the q4 analysis in §5.2 still pools nonces into the null, and the canonical-prompt test that bears on this choice is pending (§5.4). Figure 3 (Appendix A) sketches the mechanism and the test.

4 Experimental Setup

Benchmark and harness. We use τ^2-bench (Barres et al., 2025) retail and airline in text mode, on the 85 development tasks of our frozen split (58 retail, 27 airline). The harness allows 32k prompt tokens (agent maximum model length 33,792), because an 8k truncation policy lowered scores in an earlier study (Appendix E). Decoding is greedy for agent and simulator. The endpoint is database end-state match, with premature termination (step limit, error limit, infrastructure error) scored as failure.

Models. Qwen3-4B-Instruct-2507 (q4), Qwen3-8B (q8), Qwen3-14B (q14), Qwen3-30B-A3B-Instruct-2507 (q30a3) and Qwen3-32B (q32), pinned to fixed revisions and served with the Hermes tool parser; thinking is disabled for the hybrid models. The user simulator is a fixed BF16 Qwen3-30B-A3B-Instruct-2507, shared by all agents. The main matrix runs on RTX A6000 GPUs.

Perturbation matrix. A *cell* is one (model, perturbation) configuration and a *draw* is one independent run of a cell (Table 1). The Gaussian ladder adds i.i.d. Gaussian noise to every linear layer, rescaled per layer so that $\|N\|_F = k \cdot \|\mathrm{RTN}_{W4}(W) - W\|_F$ (RTN W4, group 128, asymmetric): $k = 1$ matches the per-layer error norm of RTN-W4, and $k = 0.03$ is a thirtieth of it. Embeddings, LM head, norms and MoE routers stay in BF16; quantized cells are fake-quantized (dequantized weights served in BF16).

Deterministic serving. Agents are served with vLLM batch-invariant ker-

Cells	Draws / cell	Role
BF16, canonical prompt	1	reference run
BF16, eager execution	1	agent-only numerics null
BF16, nonces 1–5	5	nonce null
Gaussian, $k \in \{0.03, 0.1, 0.3, 1, 3\}$	2 seeds; 5 at k=0.03 (q4, q8)	dose ladder; k=0.03 is the weight-only null
RTN W8 / W4 / W3 (g128, asym.)	2 (nonces 1, 2)	quantization
GPTQ W4 (calib. draws d1, d2), W3	2 per draw	quantization
AWQ W4	2	quantization
BF16 pair, 20 tasks, two nodes	1 pair	$D \equiv 0$ check

Table 1: Perturbation matrix per model. Quantized cells are run under nonces 1 and 2; Gaussian cells under the canonical prompt with different noise seeds. Completion counts per cell are in Appendix B.

nels, the FlashInfer sampler disabled, and `TORCHINDUCTOR_DETERMINISTIC=1`. Every server start runs a numerics canary against stored reference outputs for the agent and the simulator. Cross-node determinism pairs are byte-identical in prompts and completions: 257/257 decisions for q4 and 303/303 for q30a3. §6 describes the incident that led to this configuration.

Two passes: legacy and clean. A first pass (*legacy*) was served before the deterministic-Inductor fix, so each server drew one of several numerics variants at compile time; only q4 BF16 and its nonce nulls are verified to share one. The full matrix was re-run under the fix (*clean*), and every number is labelled legacy or clean. Where we pool the two, we treat the numerics variant as one more exchangeable perturbation, justified by P1 for cells above the floor; since this rests on P1 itself, pooled numbers are marked as such.

Measurements. From the decision logs we compute (i) the first divergence per task against the reference: the first decision whose raw tokens differ on an identical context (tasks ending before divergence are censored), with Kaplan–Meier (KM) estimates of $P(\text{first divergence} \le s)$; (ii) the *per-decision hazard*, the number of first divergences divided by the number of decisions at risk, with a task-bootstrap CI (because episodes last about 12 decisions, KM $P(\le 12)$ is already near 1 at the floor hazards of Table 2 and cannot rank perturbations; the hazard can); (iii) identical-context flip rates for text and tool turns and the closed-loop shift $|d|$; and (iv) the outcome statistics of §3.4.

q4 configuration (clean)	med. $\lvert d \rvert$	hazard / decision	KM $P(\le 12)$
eager BF16 (agent-only)	0.125	0.17 [0.15, 0.20]	0.96
Gaussian k=0.03 (×5)	0.125	0.18–0.23	0.97–1.0
RTN W8 (×2)	0.125	0.21 / 0.23	1.0
nonces 1–3	0.25	0.25–0.27	0.98–1.0
Gaussian k=0.1 (×2)	0.25	0.28 / 0.29	1.0
Gaussian k=0.3 (×2)	0.5	0.42 / 0.45 [0.35–0.55]	1.0
RTN W4 (×2)	1.75–2.0	0.73 / 0.78	1.0
Gaussian k=1 (×2)	2.0–2.1	0.77 / 0.99	1.0
Gaussian k=3, RTN W3	≥ 2.6	1.0	1.0

Table 2: Clean q4: median closed-loop shift $|d|$ (bf16 grid), per-decision divergence hazard (95% task-bootstrap CI where available), and episode-level divergence. Live snapshot 2026-09-29T02:36Z.

An open-loop measurement (E3) replays each cell on the reference BF16 contexts on its own server and records per-token flips and shifts over all tokens (Appendix D).

5 Results

5.1 P1: a divergence floor, then a regime that ranks the shift

Episode-level divergence saturates; the per-decision hazard does not. On clean q4 cells, every perturbation from eager execution to RTN-W8 reaches KM $P(\le 12) \ge 0.96$, so episode-level divergence cannot distinguish them. The per-decision hazard can (Table 2).

Table 2 shows the two regimes P1 predicts. Every perturbation whose median shift equals one bf16 grid step (0.125) reaches a floor of about 0.17–0.23 per decision: a change from compiled to eager kernels, Gaussian noise at 3% of the RTN-W4 error norm, and RTN-W8. The nonce members (0.25–0.27) sit above this band, at the level of $k = 0.1$. Above the floor the hazard grows slowly, roughly $2\times$ per decade of k up to $k = 0.3$ (0.42–0.45), and then jumps once tool turns start to flip (0.73–0.78 for RTN-W4, 0.77–0.99 at $k = 1$). The same structure appears in the legacy q4 cells and, with a lower floor (0.12–0.14 per decision at k=0.03 legacy; 0.11 on the two clean seeds), in q8 (Appendix G). The q8 result comes from variant-mixed legacy cells and a partial clean set (79 tasks), so we treat it as consistent with P1 rather than as a confirmation; for clean q8 at k=0.03 (seed 1), KM $P(\le 12)$ is 0.84 [0.75, 0.93], just below the 0.85 we pre-registered for P1 but above the 0.7 falsification line.

The floor is set by text-turn ties. Reference margins are two-tiered in all four models we measured. The median minimum margin of a text turn

[PENDING: Figure 1 — per-decision hazard (95% CI) versus median $|d|$ for q4, q8, q14, q30a3, q32, with one marker per family (eager, nonce, Gaussian $k = 0.03 \ldots 3$, RTN W8/W4/W3, GPTQ W4/W3, AWQ W4) — cells `a6k2_<model>_{bf16,bf16_eager,null_n1..n5,gauss<k>_s<seed>,rtn_w<b>_n<i>,gptq_w4_d<d>_n<i>,awq_w4_n<i>,gptq_w3_n<i>}`; script `analysis/insights/iter_1/verify/v01_legacy_recompute.py a6k2`; shift axis from E3 (`scripts/a6000/openloop_d.py`)]

Figure 1: Placeholder (not yet plotted): per-decision hazard versus shift, five Qwen3 models. Data: clean-pass decision logs; shift from the open-loop E3 replay, using a statistic that does not collapse near zero (the open-loop median is 1.3×10^{-6} for q4 RTN-W4; Appendix D).

is 0.0 nats (q4, q30a3) or 0.125–0.25 (q8, q14), with 2.7–6.0 tokens per decision below 0.5 nats; for tool calls it is 4.25–7.75 nats, with 0.06–0.65 tokens below 0.5. Correspondingly, at $k \leq 0.1$, identical-context text turns flip at 0.22–0.53 per decision and tool turns at 0.02–0.08 (q4, q8), and on legacy q4 cells 72–82 of 77–85 first divergences per configuration at $k \leq 0.1$, nonce and W8 are wording or format changes.

$D \equiv 0$ gives no divergence. Under the deterministic configuration, the cross-node q4 pair has 257/257 identical-context decisions with zero flips and no divergence on 20 tasks, and the q30a3 pair 303/303; a clean q4 BF16 run and one determinism replica also agree on 257/257 decisions. Conversely, eager versus compiled BF16 on clean q4 has median $|d| = 0.125$ and KM $P(\leq 12) = 0.96$. Divergence thus shows no events at $D \equiv 0$ and a hazard of at least 0.17 per decision at the smallest shift we can measure; only the episode-level KM saturates there.

Across models. Figure 1 will extend Table 2 to all five models on an open-loop shift axis. P1 holds across the family if all five models show a floor at the grid step and a hazard that separates k=0.3 from it; if any model has KM $P(\leq 12) < 0.7$ at a nonce or k=0.03, or if its KM $P(\leq 4)$ rises by more than 0.3 between k=0.03 and k=0.1, P1 is falsified for that model. On clean q4, KM $P(\leq 4)$ rises from 0.54–0.69 to 0.78–0.79 between these two values of k, below the 0.3 limit.

q4 family (draws)	Δ pp [95% CI]	wins / losses
Gaussian k=0.03 (7)	+5.1 [+1.4, +8.9]	63 / 18
Gaussian k=0.1 (4)	−1.5 [−5.1, +2.0]	27 / 18
RTN W8 (4)	+1.2 [−2.8, +5.4]	33 / 17
Gaussian k=0.3 (4)	+1.0 [−3.7, +5.7]	37 / 24
RTN W4 (4)	−2.0 [−7.6, +3.9]	33 / 24
Gaussian k=1 (4)	−2.3 [−8.7, +3.8]	41 / 35
RTN W3 (4, partial: 46 tasks)	−14.0 [−26.3, −2.4]	7 / 22
Gaussian k=3 (4, partial: 25 tasks)	−16.1 [−31.8, −0.7]	2 / 9

Table 3: q4, Δ database match against the pooled 10-draw null (pp; legacy and clean draws pooled). Wins / losses: episodes that pass a task the null majority fails / fail a task it passes (definition in Appendix A). Live snapshot 2026-09-29T02:36Z.

5.2 P2: damage appears later than divergence (tentative, Qwen3-4B)

On q4 we pool all draws of each configuration family (legacy and clean) and compare each family with a pooled null of 10 draws (canonical BF16 from both passes, nonces 1–4 legacy and 1–3 clean, and clean eager BF16), using a task-cluster bootstrap (Table 3).

Read together with Table 2, this is the two-threshold shape P2 predicts. Divergence reaches its floor at the smallest perturbation we apply (hazard 0.17, eager), while database match is not distinguishable from the null through k=1 and RTN-W4 ($\Delta = -2.3$ [−8.7, +3.8] and −2.0 [−7.6, +3.9]) and drops only at k=3 (−16 pp) and RTN-W3 (−14 pp). Non-rejection is not equivalence: we have not run a declared-margin equivalence test, and these intervals are wide. On q4 the free zone spans at least from k=0.03 to k=1 (about 1.5 decades of k); the Gaussian damage threshold lies somewhere in $(1, 3]$, a resolution of half a decade. The pre-registered pooled free-zone statistic ($k \leq 0.1$ and W8) is −0.2 pp [−5.0, +4.5] on legacy q4, −0.7 pp [−6.1, +4.6] on legacy q8, +1.9 pp [−2.6, +6.4] on clean q4 and −3.5 pp [−9.6, +1.9] on clean q8 (the last with 2–3 nulls and partial tasks). None meets the falsification rule (below −3.4 pp with a CI excluding zero), although the clean q8 point estimate is already past −3.4.

This is tentative for four reasons: (i) the damage estimate comes mostly from legacy draws (clean W3 alone: −8.5 and −9.3 pp on 39 and 25 tasks, CIs including zero); (ii) q8 shows no significant W3 damage (−3.9 pp [−13.9, +6.9]) despite tool-turn flip rates of 0.76–0.88, possibly because its lower base rate masks damage; (iii) the k=0.03 family sits *above* the null (+5.1 pp), either

chance among 11 families or a canonical-prompt effect against the nonce members (§5.4); (iv) the empty/terminated-episode audit recommended by the Agentic Benchmark Checklist (Zhu et al., 2025) has not been run on the damage cells.

What damage needs. Tool-turn flip rates are already 0.19–0.42 for RTN-W4 and 0.23–0.92 at k=1 while Δ is about -2 pp and not significant; damage appears only where tool flips approach 1.0 (k=3, W3). Most action flips are thus recovered by the loop, and the tool-turn flip rate, which our original wording of P2 named as the predictor, does not track damage. We restate the mechanism: damage requires *unrecovered flips of state-changing actions*, and the predictor to test is the write-flip rate (consistent with the 4B study, Appendix E). [PENDING: write/read/text annotation of first divergences and "undone within episode" flag (analysis V1-A4), and Spearman of Δ against write-flip rate versus tool-turn flip rate, KM $P(\leq 4)$ and median $|d|$ (P2b) — all a6k2 cells per model, E3 flip rates from `scripts/a6000/openloop_d.py`] Process channels (episode length, premature endings, repeated calls) move at k=1 while success does not, and part of the W3 damage is step-limit failure (Appendix G), as failure channels do under flat scores in Jang et al. (2026).

Across models. Figure 2 will use the same shift axis as Figure 1. [PENDING: clean q4/q8 re-test of the damage threshold — cells `a6k2_{q4,q8}_{null_n1..n5,gauss3_s1,gauss3_s2,rtn_w3_n1,rtn_w3_n2,gptq_w3_n1,gptq_w3_n2}`; `verify/v03_outcomes_pooled.py` a6k2, `verify/v04_pooled_draws.py`] P2 holds on clean data if the clean damage cells have CIs below zero while the pooled free-zone Δ stays within $\pm$3.4 pp; if clean W3 and k=3 stay indistinguishable from the null, the q4 damage threshold rests on legacy draws only; if the pooled free-zone Δ falls below -3.4 pp with a CI excluding zero, the free zone is refuted.

5.3 P3: margins, the shift distribution, and whether D is sufficient

The reference margin predicts which tokens flip on identical contexts (decision-level AUC 0.79–0.97, legacy q4 and q8 cells), replicating the inherited model. An open-loop replay of q4 RTN-W4 on the clean BF16 contexts (400 decisions, 33,836 tokens) shows the per-token flip rate falling monotonically with the reference margin, from 0.57 below 0.25 nats to 0.003 above 8 nats, while a BF16 self-replay has no flips (Appendix D). Because text turns have near-zero minimum margins in all four measured models, P3 predicts that every model reaches the floor at the grid step, with a height set by its text-tie density; q8's lower floor (0.11–0.14 versus about 0.2 for q4) is consistent with its somewhat larger text-turn margins. Preregistered margin-only divergence curves for q14 and q30a3, and a test that $D(k)$ does not depend on model size (q4 and q8 have identical clean closed-loop median $|d|$ of 0.125 / 0.25 / 0.5 at $k = 0.03/0.1/0.3$), are pending (Appendix G).

[PENDING: Figure 2 — Δ database match versus median $|d|$ per family with the null band (null-draw SD and leave-one-out range) for q4, q8, q14, q30a3, q32, free zone shaded where Δ CIs include zero — cells `a6k2_<model>_{null_n1..n5,bf16,bf16_eager,gauss<k>_s<seed>,rtn_w<b>_n<i>,gptq_w4_d<d>_n<i>,awq_w4_n<i>,gptq_w3_n<i>}`; scripts `analysis/insights/iter_1/verify/v03_outcomes_pooled.py` a6k2 and `v04_pooled_draws.py`]

Figure 2: Placeholder (not yet plotted): Δ against the reference model's perturbation null versus shift, five Qwen3 models. Data: clean-pass outcome matrices; same shift axis as Figure 1.

Is D sufficient? An open question. The strongest form of P3 says that perturbations with matched D have matched closed-loop outcomes whatever their structure. On legacy q8, AWQ-W4 and RTN-W4 land near Gaussian k=1 by closed-loop shift and tool-turn flip rate, and GPTQ-W4 between k=0.3 and k=1 (Appendix G). Outcomes, however, do not all match: pooling the available q8 draws, RTN-W4 has $\Delta = -10.5$ pp [-17.3, -4.0] (7 wins, 25 losses), whereas k=1 has 0.0 pp [-7.2, $+7.7$], AWQ-W4 $+1.6$ pp [-7.4, $+10.7$] and GPTQ-W4 -0.5 pp [-6.7, $+5.6$]. On q4 there is no such gap (RTN-W4 -2.0 pp, k=1 -2.3 pp). The q8 RTN draws include variant-mixed legacy cells, and RTN-W4's 90th-percentile shift is not larger than that of k=1, so a real gap would point to the structure or sign of the error rather than its size. D-sufficiency is unresolved; a clean q8 test is pending (Appendix G).

5.4 Consequences for evaluation

The canonical score is one draw. Re-serving the same BF16 weights with the same prompt in the legacy and clean configurations changes the q4 canonical score from 0.412 to 0.329 (-8.2 pp;

21/85 tasks = 25% discordant; McNemar $p = 0.19$) and the q8 score from 0.316 to 0.380 (+6.4 pp; 21/79 = 27%; $p = 0.38$). The pair also changed the simulator's serving (its first message differs in 11 of 20 determinism tasks), so it is an *agent + simulator* serving null. Agent-only numerics changes also flip outcomes: 4/20 tasks in the legacy cross-node pair (0.45 vs. 0.55) and 11/69 (16%) for eager vs. compiled execution on clean q4. Consistent with the Bernoulli account (§3.4), the null-draw SD is 2.9 pp on q4 (6 near-complete draws; predicted 3.2) and 3.75 pp on q8 (4 draws; predicted 3.26), ratios of 0.9 and 1.15. Canonical-prompt draws are not systematically above the other null members (q4 +1.8 pp [−2.9, +6.6]; q8 +1.9 pp [−3.0, +6.8]). These results are for q4 and q8, greedy decoding, τ^2-bench retail and airline.

Reference-only tests over-flag. A naive McNemar test against the canonical run flags legacy q4 Gaussian k=0.3 seed 1 at $p = 0.021$, whereas its null-referenced Δ is −8.3 pp [−16.3, 0.0]. The calibration result of §1 is the related single-draw failure: +0.121 with a one-sided $p = 0.0195$ that became −0.035 under three draws. In the damage cells, by contrast, pass sets are nested: cell-only wins against the canonical run are 0/10 and 0/8 for legacy RTN-W3 and 0/4 and 0/7 for k=3, against 4–7 cell-only wins per 72 tasks in free-zone cells. The clean RTN-W3 draws do not show this yet (3 versus 4 and 3 versus 3 on 39 and 25 tasks), so the cell-only win rate is a promising damage signature that is not yet confirmed on clean data. [PENDING: cell-only win rate against the null's own win-rate band (P4e) on clean damage cells — `a6k2_{q4,q8}_{rtn_w3_n1,rtn_w3_n2,gauss3_s1,gauss3_s2,gptq_w3_n1,gptq_w3_n2}`; `verify/v03_outcomes_pooled.py a6k2`] [PENDING: canonical-prompt test (P4c) comparing the weight-only null (Gaussian $k = 0.03$, s1–s5, canonical prompt), the numerics null (eager) and the nonce null — cells `a6k2_{q4,q8}_{gauss0.03_s1..s5,bf16_eager,null_n1..n5,bf16}`; `verify/v04_pooled_draws.py`]

6 Robustness and Diagnostics

The determinism incident as a finding. In the legacy pass, the q4 determinism pair served on two nodes disagreed on 20/20 tasks and 113/113 identical-prompt decisions, because torch Inductor autotuned the reduction configuration of the fused residual + RMSNorm kernel at compile time and so changed the summation order, which vLLM's batch invariance does not cover (Appendix G). The effect was that of any perturbation above the floor: median $|d| = 0.125$, non-zero shifts on 59% of tokens, KM $P(\leq 4/ \leq 12) = 0.50/0.95$ over 20 tasks, and 4/20 outcomes changed ($p = 0.63$). Deterministic Inductor compilation plus a numerics canary removed it (§4). Stacks that compile kernels may be exposed to it. Harness length, simulator-side loops and a strict error budget (Jang et al., 2026) do not change our conclusions on the data available; real quantized kernels, the strict endpoint on the damage cells, and eager nulls for the larger models are pending (Appendix F).

7 Recommendations for Evaluators

(1) **Build a null from the reference model**: at least a serving-numerics variant and weight-only noise on the canonical prompt, with prompt nonces alongside; state which members are admitted (the q4 analysis pools 10 null draws from both passes; the candidate member Gaussian k=0.03 sits above the rest). (2) **Compare at the task level against the null**, with a task-cluster bootstrap pooled over domains, not per-cell McNemar tests against one canonical run. (3) **Run each compressed cell two to three times**; a single draw can reverse sign. (4) **Pin and disclose serving numerics**, for agent and simulator: batch-invariant kernels, deterministic compilation, a numerics canary at every server start, and cross-node determinism pairs. (5) **Report divergence per decision**, not per episode. Appendix H lists four further items (open-loop fidelity, process channels, nested pass sets, harness disclosure).

8 Conclusion

Under greedy decoding, any perturbation reaching the bf16 grid step re-rolls closed-loop agent trajectories, because free-text turns are full of near-ties; a compiler configuration change, a prompt nonce, 3% of the RTN-W4 weight error and RTN-W8 reach the same per-decision floor on Qwen3-4B. A single BF16 run is one draw from a distribution, and a compressed agent should be judged against that distribution. On Qwen3-4B, success is not distinguishable from the null up to the RTN-W4 scale and falls only near 3 bits (tentative); whether this holds across sizes, and whether D predicts closed-loop effects, remains open.

Limitations

The evidence has clear boundaries. We use two τ^2-bench domains and 85 tasks, so run-level standard errors are about 3 points. The pre-registered minimum detectable effect of about 3.4 pp applies to the pooled five-model free-zone test; single-family tests on one model with 2–4 draws have less power, and because we have not run a declared-margin equivalence test, a Δ whose interval includes zero shows only that the cell is not distinguishable from the null, not that it is equivalent. All decoding is greedy; under sampling, sampling noise would dominate the null and the "one draw" issue would change form. Quantized cells are fake-quantized, and real-kernel checks are pending. One fixed user simulator is used for the main matrix (the same model as the q30a3 agent); a different simulator, or simulator-side perturbations, may change null widths. All models are from one family (Qwen3). The legacy pass is variant-mixed, and several numbers pool legacy and clean draws on the assumption that the numerics variant is an exchangeable perturbation, an assumption that itself rests on P1. The clean-pass numbers are live snapshots taken while runs were in progress. The 4B deep-dive numbers were recomputed from analysis files, not from raw runs. The closed-loop median $|d|$ is quantized on the bf16 grid, and its smallest nonzero value is the grid step at which we locate the floor; no perturbation below one grid step was tested. The open-loop $|d|$ is a different quantity that we never compare numerically with it, and matched-shift comparisons are provisional until one definition is used throughout.

Open questions.

- Does the divergence floor reached at the grid step hold for all five models, and does its height follow each model's text-tie density? (§5.1, pending)
- Does the q4 damage threshold survive on clean cells alone, and does it appear on q8 and the larger models, whose base rates differ? (§5.2, pending)
- Is the q8 RTN-W4 deficit at matched shift real, and if so, what property of RTN error causes it? (§5.3, at risk)
- Is the write-flip rate the right predictor of damage, as the recovery asymmetry in the 4B study (0.21–0.35 when the diverging action is a write versus 0.40–0.80 for a read or lookup) suggests? (§5.2, pending)
- Is the $+5.1$ pp of the Gaussian $k{=}0.03$ family above the null a chance fluctuation, a canonical-prompt effect against nonces, or an effect of the mix of passes in the family and the null? (§5.2, pending)
- Does τ^2-bench's success check count empty or terminated responses as successes in our damage cells, as the Agentic Benchmark Checklist reports for τ-bench (Zhu et al., 2025)? (not audited)
- How do our 16–27% outcome flips under numerics-only changes relate to the Replay Gap (Gonuguntla, 2026), whose abstract reports no outcome flips across 359 same-model control forks? Candidate differences are the simulator in the loop, bf16 ties in free text, and the binary database endpoint; we have not tested them.

Ethics Statement

This work involves no human subjects, crowdworkers or personal data: all users are simulated by an LLM on the public τ^2-bench retail and airline tasks, and all agent models are openly released Qwen3 checkpoints. The main risk we see is misuse of the criterion as a certificate. A compressed agent that is not distinguishable from its reference's perturbation null on 85 tasks may still fail on inputs or failure channels that the benchmark does not cover, and at this sample size non-rejection does not establish equivalence. Deployment decisions for compressed agents in consequential settings should therefore not rest on this test alone. Compute use is reported in the Responsible NLP checklist.

References

Aranyak Acharyya, Carey E. Priebe, and Hayden S. Helm. 2025. Testing for LLM response differences: the case of a composite null consisting of semantically irrelevant query perturbations. *arXiv preprint arXiv:2509.10963*.

Victor Barres, Honghua Dong, Soham Ray, Xujie Si, and Karthik Narasimhan. 2025. τ^2-Bench: Evaluating conversational agents in a dual-control environment. *arXiv preprint arXiv:2506.07982*.

Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. 2015. Scheduled sampling for sequence prediction with recurrent neural networks.

In *Advances in Neural Information Processing Systems (NeurIPS)*.

Bjarni Haukur Bjarnason, André Silva, and Martin Monperrus. 2026. On randomness in agentic evals. *arXiv preprint arXiv:2602.07150*.

Teng-Ruei Chen. 2026. The integer alibi: Localizing Cross-Kernel divergence in INT8-Quantized LLM inference. *arXiv preprint arXiv:2608.13756*.

Yihang Chen, Pin Qian, Su Wang, Chong Peng, Huan Xu, Xiyang Wu, and Yiqi Sun. 2026. Noise floor audit for agent benchmarks. *arXiv preprint arXiv:2608.22331*.

Kexin Chu, Yang Zhou, and Wei Zhang. 2026. MarginGate: sparse Margin-Triggered verification for Batch-Invariant LLM inference. *arXiv preprint arXiv:2605.30218*.

Björn Deiseroth, Max Meuer, Nikolas Gritsch, Constantin Eichenberg, Patrick Schramowski, Matthias ASSenmacher, and Kristian Kersting. 2023. Divergent token metrics: Measuring degradation to prune away LLM components – and optimize quantization. *arXiv preprint arXiv:2311.01544*. NAACL according to Semantic Scholar.

Peijie Dong, Zhenheng Tang, Xiang Liu, Lujun Li, Xiaowen Chu, and Bo Li. 2025. Can compressed LLMs truly act? an empirical evaluation of agentic capabilities in LLM compression. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*.

Gaoyuan Du, Anam Nawaz Khan, Rex Zhou, Xiaoyang Liu, Deepayan Chakrabarti, Fnu Suya, and Xueping Li. 2026. Greedy decoding is not Precision-Invariant: Cross-Precision output divergence in LLM inference. *Transactions on Machine Learning Research*.

Abhinav Dutta, Sanjeev Krishnan, Nipun Kwatra, and Ramachandran Ramjee. 2024. Accuracy is not all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Ashritha Gonuguntla. 2026. The replay gap: Static evaluation of model switching in LLM agents scores the wrong world. In *Conference on Language Modeling (COLM)*.

Horace He and Thinking Machines Lab. 2025. Defeating nondeterminism in LLM inference. Thinking Machines Lab: Connectionism (blog). September 10, 2025.

David Heineman, Valentin Hofmann, Ian Magnusson, Yuling Gu, Noah A. Smith, Hannaneh Hajishirzi, Kyle Lo, and Jesse Dodge. 2025. Signal and noise: A framework for reducing uncertainty in language model evaluation. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Jiwon Jang, Kisu Yang, Heuiseok Lim, and Hyunwoo Park. 2026. Flat score, amplified failures: How the error budget masks damage in quantized LLM agents. *arXiv preprint arXiv:2607.27275*.

Sayash Kapoor, Benedikt Stroebl, Zachary S. Siegel, Nitya Nadgir, and Arvind Narayanan. 2024. AI agents that matter. *arXiv preprint arXiv:2407.01502*. Published in Transactions on Machine Learning Research according to Semantic Scholar.

Stan Koobs and Nick W. Koning. 2026. Equivalence testing with data-dependent and post-hoc equivalence margins. *arXiv preprint arXiv:2603.16213*.

Philippe Laban, Hiroaki Hayashi, Yingbo Zhou, and Jennifer Neville. 2025. LLMs get lost in Multi-Turn conversation. *arXiv preprint arXiv:2505.06120*.

Xinyu Lian, Walid Krichene, Beichen Huang, Masahiro Tanaka, Olatunji Ruwase, Li Zhang, and Minjia Zhang. 2026. Quantization inflates reasoning: Token inflation as a hidden cost of Low-Bit reasoning models. *arXiv preprint arXiv:2606.25519*.

Haiquan Lu, Zigeng Chen, Gongfan Fang, Xinyin Ma, and Xinchao Wang. 2026. Mix-Quant: quantized prefilling, precise decoding for agentic LLMs. *arXiv preprint arXiv:2605.20315*.

Riccardo Lunardi, Vincenzo Della Mea, Stefano Mizzaro, and Kevin Roitero. 2025. On robustness and reliability of Benchmark-Based evaluation of LLMs. In *Proceedings of the European Conference on Artificial Intelligence (ECAI)*.

Moran Mizrahi, Guy Kaplan, Dan Malkin, Rotem Dror, Dafna Shahaf, and Gabriel Stanovsky. 2024. State of what art? a call for Multi-Prompt LLM evaluation. *Transactions of the Association for Computational Linguistics*.

Aditi Patodiya. 2026. Same request, different answer: Quantization amplifies Cache-Induced divergence in LLM serving. *arXiv preprint arXiv:2609.04748*.

Shahzeb Qamar, Lorenz Sparrenberg, Christian Bauckhage, Baha Rababah, Carson Leung, Murat Kantarcioglu, Cuneyt Gurcan Akcora, and Rafet Sifa. 2026. Accuracy is not enough: A Divergence-Based approach to evaluate fidelity loss in quantized LLMs. *arXiv preprint arXiv:2609.07664*.

Stephane Ross, Geoffrey J. Gordon, and J. Andrew Bagnell. 2011. A reduction of imitation learning and structured prediction to No-Regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 627–635.

Amogh Singh. 2026. Certifying compressed language models: An audit and a statistical toolkit. *arXiv preprint arXiv:2608.15046*.

Akshit Sinha, Arvindh Arun, Shashwat Goel, Steffen Staab, and Jonas Geiping. 2026. The illusion of diminishing returns: Measuring long horizon execution in LLMs. In *International Conference on Learning Representations (ICLR)*.

Yalda Taheri, Mohammad Hassan Heydari, Erfan Naaman, and Afsaneh Fatemi. 2026. Small reasoning models are instruction followers in function calling. *arXiv preprint arXiv:2608.22472*.

Universeyi. 2026. Run-to-run noise floor of the published baselines and leaderboard submissions, computed from the released trajectory files. GitHub issue #540, sierra-research/tau2-bench. Opened 2026-09-12.

Yiru Wang, Anqing Jiang, Shuo Wang, Yuwen Heng, Hai Yang, Yang Chen, and Hao Sun. 2026. Do Open-Loop metrics predict Closed-Loop driving? a Cross-Benchmark correlation study of NAVSIM and Bench2Drive. *arXiv preprint arXiv:2605.00066*.

Jiuyi Xu, Qing Jin, Meida Chen, Song Wang, Yang Sui, and Yangming Shi. 2026. VLAQuantBench: Closed-Loop evaluation of Post-Training quantization for Vision-Language-Action models. *arXiv preprint arXiv:2609.25376*.

Zihao Yang, Mosh Levy, Yoav Goldberg, and Byron C. Wallace. 2026. Compared to what? baselines and metrics for counterfactual prompting. In *Conference on Language Modeling (COLM)*.

Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. 2024. τ-bench: A benchmark for tool-agent-user interaction in real-world domains. *arXiv preprint arXiv:2406.12045*.

Jiayi Yuan, Hao Li, Xinheng Ding, Wenya Xie, Yu-Jhe Li, Wentian Zhao, Kun Wan, Jing Shi, Xia Hu, and Zirui Liu. 2025. Understanding and mitigating numerical sources of nondeterminism in LLM inference. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Liyun Zhang and Jiayi Guo. 2026. AgentDiff: Meaning-Bearing rewrites trigger deeper divergence than presentation changes in LLM agents. *arXiv preprint arXiv:2605.25981*.

Manyi Zhang, Ji-Fu Li, Zhongao Sun, Xiaohao Liu, Zhenhua Dong, Xianzhi Yu, Haoli Bai, and Xiaobo Xia. 2026. QuantClaw: precision where it matters for OpenClaw. *arXiv preprint arXiv:2604.22577*.

Yuxuan Zhu, Tengjun Jin, Yada Pruksachatkun, Andy Zhang, Shu Liu, Sasha Cui, Sayash Kapoor, Shayne Longpre, Kevin Meng, Rebecca Weiss, Fazl Barez, Rahul Gupta, Jwala Dhamala, Jacob Merizian, Mario Giulianelli, Harry Coppock, Cozmin Ududec, Jasjeet Sekhon, Jacob Steinhardt, and 6 others. 2025. Establishing best practices for building rigorous agentic benchmarks. *arXiv preprint arXiv:2507.02825*.

A Statistical Details

Δ against the null. For each task, the null mean $\bar{Y}_t^{\text{null}}$ is the mean outcome over null members (leave-one-out when the tested cell is itself a null member), computed on tasks with at least three null draws. Δ is the task average of $Y_t(C) - \bar{Y}_t^{\text{null}}$, pooled over retail and airline, and its CI is from a task-cluster bootstrap that resamples tasks and keeps all runs of a task together. Pooling across cells of one family averages over draws before the task difference is taken. A per-run rule used in an earlier version of our protocol ("inside the range of five nulls") had a high per-run false-flag rate and was replaced by this statistic.

Wins and losses against the null majority. An episode (task $\times$ draw) of a family is a win if it passes a task whose null mean is below 0.5 and a loss if it fails a task whose null mean is above 0.5; tasks with a null mean of exactly 0.5 count as neither.

Minimum detectable effect. The pooled five-model free-zone test was pre-registered with an MDE of about 3.4 pp.

Bernoulli prediction for the null SD. The predicted run-level SD is $\sqrt{\overline{p_t(1-p_t)}/n}$ with p_t estimated from the null draws of each task; the observed SD uses near-complete null draws only (at least 90% of the tasks with a null mean). The observed-to-predicted ratio was pre-registered to lie in $[0.5, 2]$. With 4–6 draws this check has little power. [PENDING: P4a pooled over five models $\times$ five nonce nulls (20 df) — `a6k2_<model>_null_n1..n5`; `verify/v04_pooled_draws.py`]

Kaplan–Meier and hazard. First divergence is the first decision whose raw tokens differ from the reference on an identical context (wording and format changes count; a behaviour-only divergence curve is a secondary measure). Tasks that end before diverging are censored. The per-decision hazard is the number of first divergences divided by the number of decisions at risk, with a task-bootstrap CI.

Calibration test of §1. The gap was tested with an exchangeable-null permutation (each task's pair of outcomes swapped with probability 0.5, 100,000 draws), which equals the exact one-sided McNemar binomial test on 8 versus 1 discordant

tasks ($p = 0.01953$); the two-sided McNemar p is 0.0389.

Task t (same instructions, same simulator) → BF16 reference / null member (nonce, Gaussian k=0.03, eager numerics) / compressed cell (RTN, GPTQ, AWQ) → decisions identical until the first flipped near-tie → free-text turn, margin below shift: flip, trajectory re-rolled; tool-call turn, margin of several nats: same action → task outcome, one Bernoulli draw with rate p_t → cell outcomes exchangeable with null outcomes? yes: perturbation-equivalent; no (nested pass set, Δ below the null band): damage.

[Diagram to be drawn from this flow.]

Figure 3: Placeholder for a conceptual schematic of the closed-loop re-roll and the equivalence test (§3). It illustrates the mechanism and is not drawn from a specific trajectory; the two-tier margin structure it depicts is measured in §5.1. Source: flowchart in the Markdown draft of this paper.

B Configuration Details

- Agent serving: vLLM with `VLLM_BATCH_INVARIANT=1`, `VLLM_USE_FLASHINFER_SAMPLER=0`, `TORCHINDUCTOR_DETERMINISTIC=1`; 32,768-token prompt budget, agent maximum model length 33,792.
- Tensor parallelism: q30a3 and q32 on two GPUs, restricted to NVLink-connected pairs after a cross-switch pair hung at NCCL initialization.
- Nonces: five fixed strings inserted as a session-identifier line in the system prompt.
- Gaussian noise: per linear layer, $\|N\|_F = k\,\|\mathrm{RTN}_{W4,g128,\mathrm{asym}}(W) - W\|_F$, $k \in \{0.03, 0.1, 0.3, 1, 3\}$.
- GPTQ and AWQ calibration sets are drawn from the model's own BF16 trajectories (trajectory-based arm): GPTQ W4 d1 and d2 use calibration draws c1 and c2; AWQ W4 and GPTQ W3 use c1.
- Models: Qwen/Qwen3-4B-Instruct-2507, Qwen/Qwen3-8B, Qwen/Qwen3-14B, Qwen/Qwen3-30B-A3B-Instruct-2507 and Qwen/Qwen3-32B, each pinned to a fixed Hugging Face revision.

[PENDING: per-cell completion counts for Table 1 — all a6k2 cells, from `docs/a6000/QUEUE.md` and `runs/a6k2_*/<domain>/status.json`; at the last snapshot 51/109 a6k2 run-domains were complete (q4 nearly complete, q8 partial, q14/q30a3/q32 calibration and determinism only)]

[PENDING: full per-cell configuration and completion table — all cells, from `scripts/a6000/make_tasks.py` and `runs/a6k2_*/<domain>/{config,status,metrics}.json`]

C Comparison with the Closest Work

Table 4 summarizes the design differences discussed in §2. Counted in episodes, Flat Score runs more episodes per cell than we do; our difference is the null and the measured margins, not the sample size.

D Open-Loop Shift Measurement (E3 Pilot)

Closed-loop estimates of D are biased, because shifts are only observed on the unflipped prefix of each decision. The open-loop measurement replays the perturbed model on the clean BF16 q4 contexts (400 decisions, 33,836 tokens). Table 5 reports the q4 RTN-W4 pilot and the BF16 self-replay control.

Over all tokens, RTN-W4 flips 6.4% of tokens and shifts 30.2% of them; 60.8% of decisions contain at least one flipped token, and the first token flips in 19.3% of decisions. The flip rate falls monotonically with the reference margin, from 0.57 below 0.25 nats to 0.003 above 8 nats. The BF16 self-replay has no token flips in any bin and $|\Delta\text{logprob}| \le 5 \times 10^{-5}$ on every token (99th percentile 4.9×10^{-5}), the rounding of the logged log-probabilities; the replay path therefore adds no perturbation beyond logging precision.

[PENDING: plot of Table 5 (flip rate versus margin bin, RTN-W4 against BF16) — to be rendered from `analysis/openloop/a6k2/q4/{rtn_w4,bf16}.json`; extend to all cells once E3 finishes]

Figure 4: Placeholder (not yet plotted). Open-loop flip rate by reference margin, q4 RTN-W4 versus BF16 self-replay. Data: Table 5.

A caution on units. The open-loop median $|d|$ over all tokens is near zero for RTN-W4 (1.3×10^{-6}; 90th percentile 0.29; 99th percentile 3.76), because most tokens have very large margins and near-certain probabilities. The closed-loop $|d|$ of Table 2 and of the §5.3 placements is a different quantity: it is computed on the unflipped prefix

Work	Closed-loop agent	Compression	Runs / episodes per cell	Null / equivalence margin	Logit margins measured
ACBench (2025)	partly (tool use, workflows)	GPTQ, AWQ, SQ, pruning	1 run	none	no (single-step diagnostics)
Flat Score (2026)	yes (τ^2-bench)	FP8, AWQ-INT4	4 trials per task (456 episodes per cell)	TOST ± 7.5 pp; noise floor from one repeat run	no
Noise Floor Audit (2026)	no (BFCL, single turn)	none	multiple reruns	reruns vs. prompt perturbations	no
Singh (2026)	no (static)	various	n/r	paired equivalence test at a declared margin (± 2 pp example)	no
Replay Gap (2026)	yes (SWE agents)	model swap; FP8/AWQ controls	control forks	same-model control forks	no
Patodiya (2026)	multi-turn tool episodes	two engines, four weight formats	n/r	none	no
Du et al. (2026)	no (single turn)	BF16 vs. FP16	n/r	none	yes (per-step model)
This work	yes (τ^2-bench, fixed simulator)	RTN W8/W4/W3, GPTQ W4/W3, AWQ W4, Gaussian ladder	2 draws $\times$ 85 tasks (170 episodes); up to 7 draws per pooled family	reference model's own perturbation null (10 draws on q4)	yes (per token)

Table 4: Comparison with the closest work. "Null" means a reference distribution against which the compressed or perturbed model is judged. Entries are taken from our literature notes; "n/r" marks cells we could not confirm in the parts we read.

BF16 margin (nats)	tokens	W4 flip	W4 med. $\lvert d\rvert$	BF16 flip
[0, 0.25)	265	0.574	0.651	0.0
[0.25, 0.5)	480	0.556	0.559	0.0
[0.5, 1)	865	0.462	0.433	0.0
[1, 2)	1,633	0.337	0.264	0.0
[2, 4)	2,557	0.196	0.115	0.0
[4, 8)	3,645	0.063	0.011	0.0
[8, ∞)	24,387	0.003	0.0	0.0

Table 5: q4, open-loop per-token flip rate by reference BF16 margin on the contexts of the clean canonical BF16 run, for RTN-W4 and a BF16 self-replay. The median $\lvert d\rvert$ column is open-loop $\lvert d\rvert$ (E3), a different quantity from the closed-loop $\lvert d\rvert$ of Table 2. Source: open-loop replay outputs for q4 RTN-W4 and BF16.

tokens of identical-context decisions and on the bf16 log-probability grid. We do not compare the two numerically, and matched-D comparisons must use one definition throughout. [PENDING: E3 across all cells — `analysis/openloop/a6k2/<model>/<cell>.json` from `scripts/a6000/openloop_d.py` (`make_tasks.py--onlye3`); report p50, p90 and the full CDF of $\lvert d\rvert$ per cell on a single definition]

E The 4B Deep Dive

These results come from an earlier study of Qwen3-4B-Instruct-2507 on two RTX 4080 GPUs: 18 GPTQ W4 (g128) checkpoints from six calibration strategies $\times$ three calibration draws, plus RTN and BF16, on the same 85 tasks, with an 8k harness and the same 4B model as the simulator. We report it separately because its harness and simulator differ from the main setup. The numbers were recomputed from analysis outputs of that study; the raw runs are not available to us for re-checking.

- *Calibration strategies are indistinguishable in closed loop.* The 18 checkpoints range from 0.271 to 0.424 in pooled database match against BF16 at 0.376 (long tasks: 0.054–0.270 against 0.297). An arm-label permutation test on the 18 checkpoint scores gives $p = 0.683$ (pooled) and 0.368 (long tasks); a draw-label permutation gives 0.881 and 0.744. Every arm-versus-arm bootstrap interval crosses zero. The calibration draw alone moves one arm's long-task success from 0.054 to 0.243. RTN scores 0.271 pooled, equal to the lowest GPTQ checkpoint.

- *Open-loop NLL is unrelated to closed-loop success.* Across the 18 checkpoints, the Spearman correlation between teacher-forced NLL per token and pooled success is -0.124 (mean over decisions) or $+0.054$ (token-weighted); both give $\lvert\rho\rvert \leq 0.13$, and they disagree in sign, so we report both. The arm with the best NLL (0.0769) and near-best full tool-call agreement (0.840) has the worst long-task success (0.117).

- *Flips are state-determined.* The reference model's NLL on a state predicts open-loop flips of compressed checkpoints there with AUC 0.93, and flip indicators correlate 0.548 within an arm and 0.538 across arms. In 0.844 of diverging (checkpoint, task) pairs, the first divergence falls on the least-confident step so far, against 0.292 by chance.

- *Failures are write-dominated and loops are mostly agent-side.* The arm $\times$ failure-category test gives $p = 0.94$. Of 150 premature terminations, 36.7% are agent-side loops, 35.3% show no repetition, 20.0% are mixed and 8.0% simulator-side; excluding simulator-side loops changes any checkpoint's score by at most 0.009.

- *Write flips are recovered less often.* Among BF16-solved tasks, the compressed checkpoint still solved the task in 0.21–0.35 of cases when its action at the first tool-turn divergence was a write, against 0.40–0.80 when it was a read or lookup (recomputed from analysis files; raw-run check pending).

- *Re-rolls dominate the W4 effect.* The net W4 loss is 0.044 [−0.038, 0.126] against gross per-task churn of 0.261; checkpoint identity explains a variance share of 0.008 against a residual of 0.490; and the gain rate (fail under BF16, pass under W4) is 166/954 = 0.174, so W4 pass sets are not nested in BF16's.

Taken together with the main results, these describe a regime in which W4 compression of a 4B agent is not distinguishable from re-rolling, and in which comparisons of calibration strategies against one reference run mostly measure re-rolls.

Harness length. Table 6 compares the 8k truncation harness of the earlier study with the 32k harness on the same 85 tasks (Qwen3-4B, BF16 and the first calibration draw of each arm). Removing truncation raises every model by 0.01–0.04 and leaves the ranking nearly unchanged. One W4 checkpoint (MatchedAlt c1) scores above BF16 under both harnesses, which is itself an instance of re-roll noise.

Model	8k	32k	8k trunc. retail / airline
BF16	0.376	0.412	0.29 / 0.20
Static c1	0.365	0.376	0.56 / 0.22
FP-Traj c1	0.282	0.306	0.53 / 0.50
Traj-All c1	0.318	0.353	0.58 / 0.33
Traj-Novel c1	0.318	0.353	0.53 / 0.34
MatchedRandom c1	0.329	0.341	0.49 / 0.36
MatchedAlt c1	0.424	0.447	0.53 / 0.45

Table 6: Pooled database match under 8k and 32k harnesses, with the 8k truncation rate per domain (4B study). Traj-All and Traj-Novel are two trajectory-based calibration arms, renamed for anonymity.

F Further Robustness Checks

Simulator-side loops. Scoring simulator-side loops as missing rather than as failures moves any checkpoint's success by at most 0.009 (Appendix E), so attributing premature endings to the agent does not drive our conclusions.

Strict error budget. Following Jang et al. (2026), we also score a strict endpoint that fails episodes with more than two tool errors. On the legacy q4 pilot this changes Δ by at most 1.7 pp. [PENDING: strict-budget endpoint and empty-episode audit on the damage cells — `a6k2_{q4,q8}_{gauss3_s*,rtn_w3_n*,gptq_w3_n*}`; `analysis/insights/iter_1/s06_pilot_delta.py` (E4c)]

Real quantized kernels. Our quantized cells use fake quantization. The Replay Gap result (Gonuguntla, 2026) suggests that null width can depend on the serving format. [PENDING: Marlin real-kernel determinism pairs for q8 awq_w4 and gptq_w4 (E7), and real-format versus fake-quant divergence and success for gptq_w4_d1 on q8/q32 — Phase 4 cells; `scripts/a6000/make_tasks.py`] If real-kernel reruns are byte-identical, the null structure of §3.4 carries over unchanged; if they are not, the per-format rerun null must be measured and reported per arm.

Agent-only numerics null for all models. [PENDING: bf16_eager cells for q14, q30a3, q32 (V1-A3) — `a6k2_{q14,q30a3,q32}_bf16_eager`; `verify/v01_legacy_recompute.py` a6k2 and `verify/v03_outcomes_pooled.py` a6k2]

G Additional Results

P1 in legacy cells and on q8. Legacy q4 cells show the same two regimes as Table 2: numerics variant 0.18 [0.14, 0.24] per decision; k=0.03: 0.19–0.20; k=0.3: 0.39–0.42; k=1: 0.78–0.98. Legacy q8 has a lower floor (0.12–0.14 per decision at k=0.03; 0.11 on the two clean k=0.03 seeds) with nonce 0.21, k=0.3 0.29–0.34, GPTQ-W4 0.41–0.46 and AWQ-W4 0.51–0.56. On clean cells at k=0.3 the text-to-tool flip ratio is 5.8 / 5.7

(q4) and 6.4 / 11.3 (q8).

Process channels. On legacy q4 retail at k=1, episodes grow from 12.0–12.3 decisions (nulls) to 15.1 / 16.9, premature endings rise from 0 to 0.034 / 0.069, and repeated tool calls reach 0.236 against 0.04–0.07, while RTN-W4 and every $k \leq 0.3$ cell stay at null level (11.1–12.9 decisions, no premature endings). At RTN-W3, the step limit ends 7/33 and 8/32 of finished legacy episodes (both domains) against none in the nulls, so part of the W3 damage is process failure.

Cross-model predictions. We pre-registered margin-only first-divergence curves for q8, q14 and q30a3; for example, at k=0.3 the predicted $P(\leq 0/ \leq 4/ \leq 8)$ is 0.25 / 0.77 / 0.99 for q14 and 0.35 / 0.61 / 1.00 for q30a3, to be scored within ± 0.15 after removing the survivorship bias (the margin-only prediction underpredicted legacy q4 by 0.1–0.15). [PENDING: observed first-divergence curves for q14 and q30a3 at $k = 0.3$ against the pre-registered predictions — cells `a6k2_{q14,q30a3}_gauss0.3_s{1,2}`; scripts `analysis/insights/iter_1/s03_margin_predictions.py` and `verify/v01_legacy_recompute.py a6k2`] A related prediction is that $D(k)$ does not depend on model size. On clean cells, q4 and q8 have identical closed-loop median $|d|$ at $k = 0.03/0.1/0.3$ (0.125 / 0.25 / 0.5); the P3a test must compare all models on a single $|d|$ definition (closed-loop or E3 open-loop, not a mix). [PENDING: P3a, median $|d|$ at $k = 0.1/0.3/1$ for q14, q30a3, q32 within $1.5\times$ of q4 — cells `a6k2_{q14,q30a3,q32}_gauss{0.1,0.3,1}_s{1,2}`; E3]

Matched-D placements on q8. On legacy q8, by closed-loop shift, GPTQ-W4 (median $|d|$ 0.75, 90th percentile 2.0–2.25) sits between k=0.3 and k=1, and AWQ-W4 (1.5; 3.5) and RTN-W4 (1.5; 4.1–4.25) sit near k=1 (1.625–1.75; 4.6–4.75), with matching tool-turn flip rates (AWQ-W4 0.38–0.41, RTN-W4 0.39–0.44, k=1 0.35–0.41). [PENDING: clean q8 matched-D test — cells `a6k2_q8_{rtn_w4_n1,rtn_w4_n2,gauss1_s1,gauss1_s2,awq_w4_n1,awq_w4_n2,gptq_w4_d1_n1,gptq_w4_d1_n2,gptq_w4_d2_n1,gptq_w4_d2_n2}` with E3 D (p50 and p90); `verify/v04_pooled_draws.py`, `scripts/a6000/openloop_d.py`] These placements use closed-loop $|d|$, whereas the pending test uses open-loop E3 $|d|$; the pending placement will be read on the E3 scale alone. If the RTN-W4 gap disappears on clean cells at matched p50 and p90, the matched-D claim survives on two models; if it persists, the claim narrows to "error-minimizing quantizers (GPTQ, AWQ) sit on the Gaussian band; RTN does not", and RTN would carry damage beyond its logit shift (for example through systematic rather than zero-mean error).

The determinism incident in detail. The compiled kernels of the legacy q4 determinism pair were identical across its two nodes, and on a third node cold, warm, concurrent and prefix-cache-free starts all reproduced the second node's output (30/30); eager execution produced a third variant. Inductor chose the reduction configuration of the fused residual + RMSNorm kernel by timing it at compile time (`R0_BLOCK` 2048 on one node, 4096 elsewhere). The q30a3 pair, whose autotuning happened to pick the same configuration on both nodes, was fully identical.

H Further Recommendations

Continuing the checklist of §7; (†) marks an item resting on tentative findings. (6) **Do not substitute open-loop fidelity for closed-loop success**: teacher-forced NLL and tool-call agreement did not track closed-loop success in our 4B study. (7) **Report process channels and a strict endpoint**: decisions per episode, premature endings, repeated tool calls and a strict error budget can move before the headline score. (8) **Look for nested pass sets** (†): damage shows up as a compressed cell that almost never passes a task the null fails; re-rolls do not. (9) **Disclose the harness**: context length and truncation policy moved scores by 0.01–0.04 in our setting.